



National Institute of Technology Kurukshetra

Department of Computer Engineering

Programming using Python (CSIC-104)

Branch: RA (II Semester)

Mid Term- I

Date of Exam – 20-February-2026

Time- 09:00 AM-09:50 AM [50 min]

Max Marks:15

Q 1	Given two numbers n and k such that k must be less than number of bits in n, write program to find the minimum number of flips required to maximize given number by flipping its bits such that the resulting number has exactly k set bits. Example: <table><tr><th>Input</th><th>Output</th></tr><tr><td>n = 14, k = 2</td><td>Min Flips=1</td></tr><tr><td>n = 145, k = 4</td><td>Min Flips=3</td></tr></table>	Input	Output	n = 14, k = 2	Min Flips=1	n = 145, k = 4	Min Flips=3	[5]
Input	Output							
n = 14, k = 2	Min Flips=1							
n = 145, k = 4	Min Flips=3							
Q 2	The batch representative of a batch is to be elected by an election process. The name all the students of the batch is given as the first input. The list of candidates of all the probable representatives are given as the second input. The choice of candidates is given as third input. Write a program to find out the winner with maximum votes. Example: <table><tr><th>Input</th><th>Output</th></tr><tr><td>A B C D E B C C B C B B</td><td>Winner is B with 3 votes</td></tr></table>	Input	Output	A B C D E B C C B C B B	Winner is B with 3 votes	[5]		
Input	Output							
A B C D E B C C B C B B	Winner is B with 3 votes							
Q 3	Given two Strings, where all characters are separated by a delim, write a Python code to check if both the strings contain the same characters. Example: Input: test_str1 = 'e!e!k!s!g', test_str2 = 'g!e!e!k!s', delim = '!' Output: True	[5] 5						
Q 4	Write a python program to reverse all the words in a given sentence. Example: Input: hi, how are you? Output: ih, who era uoy?	[2] (5)						